TEACHER GUIDE · GRADE 1 · UNPLUGGED SEQUENCING

Does the Order Work?

Quick facilitation notes & answer key

Goal

Students read three algorithms and judge whether the order works, explaining why, then rewrite a broken one so it works. Reading existing code and describing how its order affects whether it succeeds is C3.2; rewriting the broken order into a working one is C3.1.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: act out each algorithm so a wrong order is obvious (you can't slide before you sit at the top).

Run it (about 20 min)

1. I DO: read Algorithm A. 'You need the circle before you can add rays — so this order works.'
2. WE DO: Exercise 1 — A works (YES). Exercise 2 — B does NOT work; you can't eat before pouring the cereal and milk.
3. YOU DO: Exercise 3 — C does NOT work; rewrite it: climb the ladder, sit at the top, slide down.
4. Connect: 'All the right steps still fail if they are in the wrong order — order is part of the code.'

Answer key (modelled by the run-gate)

Ex 1 — A WORKS (YES): you draw the circle first, then add rays and colour it; each step has what it needs.

Ex 2 — B does NOT work (NO): eating is first, but you must pour the cereal then the milk before you can eat. Eating should be last.

Ex 3 — C does NOT work (NO). A working order: climb the ladder, sit at the top, slide down (you must sit at the top before you can slide down).

Watch for & differentiate

Common slip: saying an order works because all the steps are present — point out that the POSITION of a step matters, not just that it is there.

Simplify: act out Algorithm C; the wrong order becomes obvious when you try to slide before sitting.

Extend: ask students to break Algorithm A on purpose by moving one step, and explain why it no longer works.

Success indicator: the child marks A=YES, B=NO, C=NO and rewrites C as climb, sit, slide with a reason.